



Selecting single items to measure doubly concrete constructs: A cautionary tale[☆]



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ABSTRACT

Single-item measures have recently become more en vogue due to studies arguing in favor of their psychometric properties vis-à-vis multi-item scales. However, their effective use requires (1) expert raters to designate the focal construct as being doubly concrete and (2) researchers to find a good single item to represent the construct. This study examines whether expert raters identify the doubly concrete nature of constructs that prior research presents as exemplary in this respect. Furthermore, the study compares the efficacy of a broad range of selection mechanisms based on expert judgment and statistical criteria for identifying the best item in a scale. The results show that expert raters do not share the commonly held belief that researchers can validly measure constructs such as attitude toward the ad, or brand, with single items. Further analyses show that neither rater assessments nor statistical criteria prove valuable regarding identifying an appropriate single item from a set of candidate items.

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1. Introduction

The extent to which successive generations of researchers and practitioners raise an issue is a subtle indicator of its importance. The benefits and limitations of single-item (SI) versus multi-item (MI) measures is one such issue that scholars have heatedly debated across a variety of disciplines as diverse as health care (e.g., Dolbier, Webster, McCalister, Mallon, & Steinhardt, 2005), sports management (Kwon & Trail, 2005), organizational psychology (e.g., Nagy, 2002), and marketing (e.g., Bergkvist, 2015, Bergkvist & Rossiter, 2007, 2009, Diamantopoulos, Sarstedt, Fuchs, Wilczynski, & Kaiser, 2012, Kamakura, 2015). Bergkvist and Rossiter (2007, 2009), who question the universal usefulness of MI measurement on theoretical and empirical grounds, produced two of the most prominent studies in this stream of research. Drawing on Rossiter's (2002, 2011) C-OAR-SE procedure, the authors argue that if the object of the construct (e.g., a brand, or an ad) is concrete and singular, and if its attribute (e.g., an attitude, or a perception) is also concrete, using

an SI to operationalize this construct is conceptually legitimate. In support of this view, Bergkvist and Rossiter (2007, 2009) report empirical findings showing that SI and MI measures are similar in terms of predictive validity and conclude that “carefully crafted single-item measures—of doubly concrete constructs—are at least as valid as multi-item measures of the same constructs, and that the use of multiple items to measure them is unnecessary” (Bergkvist & Rossiter, 2009, p. 618). Obviously, their recommendation is very attractive for researchers because the authors offer a theoretical and empirical justification for a simpler measurement, thereby producing substantial savings in data gathering efforts and costs. Unsurprisingly, authors increasingly justify their use of SIs by citing the article of Bergkvist and Rossiter (2007) (Kamakura, 2015), which has attracted over 460 SSCI citations as of January 2016.

At the same time, however, follow-up studies cast a different light on Bergkvist and Rossiter's (2007, 2009) work. For example, Diamantopoulos et al. (2012) systematically examine the influence of different measurement and data characteristics, such as the inter-item correlations, the number of items, and the sample size, on the predictive validity of SI versus MI measures. The authors conclude that “opting for SI measures in most empirical settings is a risky decision as the set of circumstances that would favor their use is unlikely to be frequently encountered in practice” (Diamantopoulos et al., 2012, p. 446). Specifically, researchers should consider SI measures (rather than an MI scale) only when (1) working with small sample sizes (i.e., $N < 50$), (2) the expected effect sizes are small, (3) the items of the originating MI scale are highly homogeneous (i.e., Cronbach's

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alpha > 0.90), and (4) the items are semantically redundant. Similarly, Kamakura (2015) argues that Bergkvist and Rossiter (2007) confounded their results with spurious correlations due to common method biases and replicates their study taking measurement error into account. The results show that, once corrected for measurement errors, MI scales consistently outperform their SI equivalents.

Despite these discouraging results, researchers may still decide to opt for SI measures in the light of their manifold practical advantages (e.g., Drolet & Morrison, 2001; Fuchs & Diamantopoulos, 2009; Wanous, Reichers, & Hudy, 1997). In this case, however, two fundamental issues remain.

First, according to Bergkvist and Rossiter (2007, 2009), for a phenomenon to be amenable to SI measurement, a group of expert judges needs to identify the object and attribute that researchers are evaluating as being concrete. Researchers can consider an object as being concrete when virtually all the raters (e.g., respondents in a survey) “know what the object is and that, for them, there is only one object” (Rossiter, 2002, p. 311). Similarly, “a concrete attribute has virtually unanimous agreement by raters as to what it is, and they clearly understand that there is only one, or holistically one, characteristic being referred to when the attribute is posed, as in a questionnaire or interview, in the context of the to be rated object” (Rossiter, 2002, p. 313). To summarize, in order to justifiably use SI measures, a set of expert judges should agree that the construct under consideration is, in line with Rossiter's (2002, 2011) C-OAR-SE procedure, indeed doubly concrete. However, to date, no study has evaluated whether expert judges share Bergkvist and Rossiter's (2007, 2009) notion of the doubly concrete nature of such constructs as attitude toward the ad, or attitude toward the brand. The authors merely state that “according to expert judgment based on the C-OAR-SE procedure, A_{Ad} (or L_{Ad}) and A_{Brand} [i.e., attitude toward the ad, likeability of the ad and attitude toward the brand] are two such constructs [i.e., doubly concrete constructs]” (Bergkvist & Rossiter, 2007, p. 176) without, however, providing any information on the number and qualification of the expert judges involved, or the procedure(s) used to elicit expert agreement. As noted elsewhere, “JR [John Rossiter] ignores his own advice” (Diamantopoulos, Sigauw, & Cadogan, 2008, p. 390).

Second, once researchers opt for SI measurement, they still need to decide on a particular item to operationalize the construct. Unfortunately, prior research has not reached consensus on this issue. For example, Rossiter (2002, p. 310) recommends using expert raters—that is, “[a] small group of judges with expertise regarding the construct”—to select the SI. Other researchers (e.g., Netemeyer, Bearden, & Sharma, 2003) recommend the use of actual respondents (i.e., naive raters) for scale reduction purposes who—unlike expert raters—offer an intuitive judgment without domain-specific background. Yet another approach based on judgment involves the creation of a free-standing, global SI on the grounds of content validity considerations (e.g., Bergkvist & Rossiter, 2009; Wanous et al., 1997). Finally, authors suggest using statistical criteria, such as indicator loadings (e.g., Loo, 2002; Sarstedt & Wilczynski, 2009). Methodological studies that empirically compare the validity of SI and MI measures considered several of these options (see Table 1). Some authors choose the item with the highest loading (e.g., Loo, 2002), others rely on expert raters (e.g., Sarstedt & Wilczynski, 2009) or create a global SI (e.g., Bergkvist & Rossiter, 2009), and still others state no rationale at all (e.g., Bergkvist & Rossiter, 2007). However, none of these authors offers a compelling argument as to why they favor a particular approach (or why they reject alternative approaches). Most importantly, no study has yet compared the efficacy of the different approaches to SI selection.

Against this background, the purpose of this study is twofold. First, the study assesses whether the key constructs that Bergkvist and Rossiter (2007, 2009) examine in their studies as prototypical of SI measurement can be considered doubly concrete. Following the guidelines in Rossiter's (2002, 2011) C-OAR-SE procedure, the analysis draws on expert judgment to ascertain whether, or not, the focal constructs

have “a simple, clear object (e.g., an ad or a brand) and a single and single-meaning attribute (e.g., liking)” (Bergkvist, 2015).

Second, the study compares the efficacy of different SI selection approaches in finding the best SI; that is, the one with the highest predictive validity. While other types of validity (particularly discriminant validity) are also relevant when developing/selecting a measure, the current focus is on predictive validity because this criterion is particularly important for decision-making purposes (e.g., Aaker, Kumar, & Day, 2010; Crocker & Algina, 2008) and authors used it widely in prior empirical research comparing SI and MI measures (e.g., Bergkvist & Rossiter, 2007, 2009; Diamantopoulos et al., 2012; see also Table 1). Note that the item with the highest predictive validity does *not* necessarily match—let alone outperform—the predictive validity of an MI scale (Diamantopoulos et al., 2012). Using an SI is, more often than not, likely to be a second best solution, which, however, may be tolerable in light of the practical benefits of SIs (Fuchs & Diamantopoulos, 2009). Thus, the present study does *not* aim at making a case for SIs and acknowledges that the development of new measurement instruments arguably benefits from a MI operationalization (e.g., DeVellis, 2003; Netemeyer et al., 2003). SIs are also not suited for testing complex theories involving causal relationships and structural equation modeling (Hair, Sarstedt, Ringle, & Mena, 2012). In short, using a SI measure implicitly assumes that time- and cost-saving considerations take precedence over a more comprehensive, and thus generally more precise, measurement. The reality, however, is that authors *do* use SI measures widely in empirical research and they are therefore worthy of attention.

An evidence-based decision as to which SI a researcher should use requires an empirical investigation. However, evidently, comparing the performance of alternative SI measures relative to each other and against an MI measure runs contrary to the very idea of limiting data collection to just one item for the construct of interest. In any given study, administering an MI first and *then* selecting an SI would be nonsensical. A more reasonable approach is to disentangle SI selection and SI application in a two-stage procedure. In the first stage, an SI selection study investigates the performance of alternative SIs and, thus, informs subsequent application studies that administer a specific SI in substantive research endeavors. An SI selection study corresponds to a scale development project, which is also much more comprehensive than a scale's subsequent application in practice. However, published evidence related to existing instruments may allow for conclusions in terms of potentially suitable SIs, provided that SI selection studies converge on specific SI selection criteria that are generalizable and included in typical scale development reports.

2. Methods and analysis

2.1. Overview

The study uses data from consumers' responses to advertisements in order to assess the predictive validity of attitude toward the ad (A_{Ad}) on attitude toward the brand (A_{Brand}), as well as A_{Brand} on brand purchase intention (PI_{Brand}). According to Rossiter's (2002, 2011) C-OAR-SE procedure, A_{Ad} , A_{Brand} , and PI_{Brand} are all doubly concrete constructs and should thus be measured with SIs (Bergkvist & Rossiter, 2007, 2009). Similarly, Bergkvist (2015) explicitly states that “examples of doubly concrete constructs include attitude toward the ad (A_{Ad}), brand attitude (A_{Brand}), and brand purchase intention (PI_{Brand}).”

The analysis first centers on the question whether expert raters perceive the constructs under consideration as doubly concrete and, therefore, are amenable to SI measurement. Next, the analysis examines whether different SI selection approaches converge to the same item. The approaches include (1) judgment-based methods using expert raters, naive raters, and global SIs; (2) criteria based on classical test theory; and on (3) item response theory. Finally, the analysis evaluates whether the SIs that the selection approaches above identified are the ones with the highest predictive validity (i.e., the “best” items). Clearly,

Table 1
Review of Methodological Studies Involving SI Selection.

Study	SI selection approaches									Validity type	Focal construct(s)	SI performance
	n.s.	Expert raters	Naive raters	Global SI	Item-to-total correlations	Factor loadings (PAF)	Factor loadings (CFA)	Squared multiple correlations (CFA)	Rasch analysis			
Bergkvist (2015)	X									Predictive	A _{Ad} , A _{Brand}	+
Bergkvist and Rossiter (2007)	X									Predictive	A _{Ad} , A _{Brand}	+
Bergkvist and Rossiter (2009) ^a				X						Predictive	A _{Ad} , A _{Brand} , PI _{Brand}	+
Christophersen and Konradt (2011)				X						Convergent	Assessment of online store usability	+
Diamantopoulos et al. (2012) ^a										Predictive	A _{Ad} , A _{Brand} , hedonic/utilitarian attitudes toward the product	–
Dolbier et al. (2005)				X						Predictive	Job satisfaction	+
Gardner et al. (1998)				X						Convergent, discriminant	Attention at work	–
Kamakura (2015)	X									Concurrent/predictive	A _{Ad} , A _{Brand}	–
Keiningham, Cooil, Aksoy, Andreassen, and Weiner (2007)	X									Predictive	Satisfaction, word-of-mouth, repurchase intention	0
Kwon and Trail (2005)				X						Predictive	Team commitment, team identification	+
Loo (2002)					X	X				Convergent	Belief in a just world	–
Nagy (2002)				X						Convergent, predictive	Job satisfaction	+
Robins Fein, Barton, and Green (2001)				X						Convergent, predictive	Self-esteem	0
Ruekert and Churchill (1984)				X						Convergent	Channel member satisfaction	0
Sarstedt and Wilczynski (2009)		X				X				Predictive	Customer satisfaction/loyalty	–
Wanous et al. (1997)				X						Convergent	Job satisfaction	+
This study		X	X	X	X	X	X	X	X	Predictive	A _{Ad} , A _{Brand}	–

A_{Ad} = attitude toward the ad; A_{Brand} = attitude toward the brand; PI_{Brand} = brand purchase intention; n.s.: not specified; +: results favor SI measures; –: results favor MI measures; 0: mixed results.

^a Study considers all items in the scale as single items.

confidence in a certain item would be enhanced if multiple approaches identified the *same* item as being suitable for serving as the SI measure.

2.2. Participants

The first sample, which is the basis for determining expert raters' evaluation of the doubly concrete nature of the focal constructs, comprises 50 eminent marketing scholars whom the authors invited via e-mail to participate in "a study about marketing." Of the 23 respondents who fully completed the survey, the analysis follows Diamantopoulos (2005) and only considers those who had at least 3 years of experience in the substantive area (i.e., advertising/branding) and methodological expertise (i.e., in scale development and measurement theory). Furthermore, respondents had to agree to one opinion leader statement taken from Reynolds and Darden (1971), which identified them as proficient in the respective field. Applying these criteria generated a final sample of 13 expert raters.

The second sample of naive raters draws on raw data from Study 1 in Diamantopoulos et al. (2012). In the latter, the authors sent e-mails to a total of 2200 randomly chosen undergraduate students from all fields at a major university, inviting them to participate in the study. 520 respondents completed the survey fully.

2.3. Measures

For construct measurement, this study draws on Bergkvist and Rossiter's (2007, 2009) seven-point semantic differential scales, which many prior studies also used (e.g., MacKenzie & Lutz, 1989; Mitchell, 1986). Similarly, this study adopts Bergkvist and Rossiter's (2009)

bipolar seven-point global (tailor-made) SI measures of A_{Ad} and A_{Brand}. These latter items precede the MI scales to avoid priming effects and contamination by the scale (Bergkvist & Rossiter, 2009; DeMoranville & Bienstock, 2003). The attribute descriptions, response alternatives, coding and labeling are analogous to Bergkvist and Rossiter (2007, 2009).

Factor analysis results on the MI measures provide support for the scales' unidimensionality since all items load highly on a single, main factor. The coefficient beta values range from 0.55 (A_{Ad} (pain relievers)) to 0.82 (A_{Brand} (insurance)) and are thus above the common threshold of 0.50 (Revelle, 1979). The Cronbach's alpha coefficients range between 0.87 (A_{Ad} (pain relievers)) and 0.88 (A_{Brand} (jeans)), thus providing support for the scales' reliability (Nunnally & Bernstein, 1994).

2.4. Procedure

Expert raters saw two of a set of four ads and short definitions of the focal predictor constructs, A_{Ad} and A_{Brand} (Stanton, Sinar, Balzer, & Smith, 2002). The ads match the four quadrants in the Rossiter–Percy grid in order to account for potentially differing strengths of construct relationships across the quadrants (Bergkvist & Rossiter, 2007, 2009; Rossiter, Percy, & Donovan, 1991). Following Bergkvist and Rossiter (2007, 2009), the products depicted in the ads were insurance, jeans, pain relievers, and coffee. The advertisements were original ads from foreign countries, so that the respondents knew neither the brand nor the advertisement. Expert raters indicated whether the A_{Ad} and A_{Brand} constructs are doubly concrete by using Rossiter's (2002) expert judges rating form (i.e., whether they believed that an SI was sufficient to measure the constructs under consideration). Furthermore, expert raters

selected the SI that, in their opinion, best represents each construct if only one item were used.

Naive raters underwent the same procedure. However, contrary to the expert raters, the naive raters did not comment on the doubly concrete nature (or otherwise) of the focal constructs. Instead, they indicated their brand purchase intentions (PI_{Brand}) to enable the assessment of predictive validity.

3. Results

3.1. Expert assessment of the construct nature

The first step in the analysis involves examining the evaluations of expert raters and naive raters (Table 2). The vast majority of expert raters do *not* believe that A_{Ad} and A_{Brand} are doubly concrete. More precisely, only one out of 13 expert raters indicates that these constructs can be validly measured with SIs. Therefore, Rossiter's (2002, 2011) and Bergkvist and Rossiter's (2007, 2009) arguments regarding the concrete nature of these constructs are not mirrored in experts' opinions on the adequacy of SI measures for capturing these constructs.

While these findings generally advise against the use of SIs according to C-OAR-SE, the fact that experts do not agree on the double-concreteness still does not mean that the constructs could not—at least in principle—be amenable to SI measurement. Particularly with global SI questions, respondents consider all the aspects of and individual preferences for the construct being measured (Nagy, 2002) and thus provide a more tailor-made picture of the focal construct. As Youngblut and Casper (1993, p. 459) note, SIs “allow the subject to take personally salient features of the situation into account when providing a response.” That is, with global SI measures, respondents tend to ignore aspects that are not relevant to their situation and weight the relevant aspects differentially to provide a single rating (De Boer, van Lanschot, de Haes, & Sprangers, 2004). This is particularly true of measures of highly complex phenomena, such as corporate reputation (e.g., Sarstedt, Wilczynski, & Melewar, 2013), and time perspectives (e.g., Kuppelwieser & Sarstedt, 2014a, 2014b), which generally require

many items to capture each relevant dimension or aspect. The question now becomes *how* to identify a suitable SI for construct measurement.

3.2. Single-item selection

Selecting an SI from the A_{Ad} and A_{Brand} scales is based on the following approaches: (1) the expert raters' assessment, (2) the naive raters' assessment, (3) the maximum item-to-total correlations (ITC), (4) the maximum item loading derived from the principal axis factoring (PAF), (5) the maximum item loading derived from a one-factor congeneric confirmatory factor analysis (CFA) model, (6) the maximum squared multiple correlations (SMCs) based on a CFA, and (7) the minimum difference between the location of the item thresholds and the distribution of the respondents based on the polytomous Rasch model within the item response theory framework (see Appendix A for further details). In addition, the analysis considers (8) a global (tailor-made) SI as an alternative to choosing an item from an existing MI scale.

Table 2 shows that experts' and naive raters' assessments show considerable variability. Naive raters do not achieve consensus—especially regarding A_{Brand} —on which item to select. Overall, expert (naive) raters achieve Fleiss's (1971) Kappa values of 0.39 (0.38) for A_{Ad} and 0.40 (0.32) for A_{Brand} , suggesting only moderate agreement on item choice. Finally, comparing the expert and naive raters' assessments reveals that the two rater groups differ in their SI preferences. Experts advocate the *like/dislike* item for A_{Ad} and A_{Brand} , whereas naive raters prefer *pleasant/unpleasant*.

Next, ITCs, as well as factor loadings (both from PAF and CFA) and squared multiple correlations of A_{Ad} and A_{Brand} identify the SI that has the strongest relationship with the underlying predictor construct in each of the product categories. The results show some variation in SI selection, depending on the product category involved (Table 3). For example, selection for A_{Brand} based on the highest loading from PAF (Table 3B), suggests that *good/bad* is the best item in the coffee category, *like/dislike* is best in the jeans and pain relievers categories, whereas *pleasant/unpleasant* is best in the insurance category. Similarly, CFA-based criteria indicate three different items for measuring A_{Ad} , depending on the product category (Table 3A). Overall, *within each* classical test theory-based approach, the chosen items always differ between at least two product categories. However, the comparison of the choice patterns across different classical test theory-based approaches reveals only very limited variation. For example, for A_{Brand} , all approaches point at the same item in the jeans, pain relievers (i.e., *like/dislike*), and insurance (i.e., *pleasant/unpleasant*) categories. The only exception is the coffee category, where the loading from PAF suggests *good/bad*, whereas the other criteria indicate *like/dislike*. The item selection for A_{Ad} shows variation with regard to ITCs and loadings (PAF) on the one hand, and CFA-based criteria on the other. Overall, however, the classical test theory-based approaches correspond closely.

Item selection based on the Rasch analysis using the software RUMM 2030 (Andrich, Sheridan, & Luo, 2012) first involves a combined fit assessment of the individual items comprising the scale, plus the tailor-made single item. Table B1 in Appendix B provides the item parameters and fit statistics. Table B2 shows the average distance of the actual thresholds from their ideal locations of all items considered as SI measures after the fit assessment. Specifically, the analysis assesses the items' suitability to be singled out as an SI measure, assuming the person distribution is normal, with the mean and standard deviation estimated from the data. Across all the product categories, the analysis shows that *like/dislike* is the best item for A_{Ad} . The only exception is the coffee category, where *favorable/unfavorable* better targets the sample of respondents. All the items in A_{Ad} , as well as the global SI, are at fairly similar spots on the latent continuum across the product categories, making them similarly suitable for SI selection. *Like/dislike* is generally the best candidate for A_{Brand} , except for the insurance category, where *pleasant/unpleasant* performs slightly better.

Table 2
Item selection (expert and naive raters).

Expert raters: $n = 13$		Is an SI sufficient?		Kappa
		Yes	No	
A_{Ad}				
Like/dislike	62% (8)	1	12	.39
Good/bad	15% (2)			
Pleasant/unpleasant	23% (3)			
Informative/uninformative	0% (0)			
Favorable/unfavorable	0% (0)			
A_{Brand}				
Good/bad	8% (1)	1	12	.40
Like/dislike	62% (8)			
Pleasant/unpleasant	31% (4)			
Useful/useless	0% (0)			
Naive raters: $n = 520$				
A_{Ad}				
Like/dislike	24% (123)	n/a	n/a	.38
Good/bad	2% (12)			
Pleasant/unpleasant	58% (299)			
Informative/uninformative	12% (63)			
Favorable/unfavorable	4% (23)			
A_{Brand}				
Good/bad	8% (42)	n/a	n/a	.32
Like/dislike	42% (216)			
Pleasant/unpleasant	42% (219)			
Useful/useless	8% (43)			

n/a: not applicable.

Table 3

SI selection based on different approaches.

A: A _{Ad} (MI and SI) → A _{Brand} (MI)							
	Item selection						
	Expert raters	Naive raters	Highest ITC	Highest loading (PAF)	Highest loading (CFA)	Highest SMC (CFA)	Rasch
A_{Ad} MI (baseline)							
Like/dislike	X		X ^{jeans, pain relievers, coffee}	X ^{jeans, pain relievers, coffee}	X ^{pain relievers, coffee}	X ^{pain relievers, coffee}	X ^{insurance, jeans, pain relievers}
Good/bad			X ^{insurance}	X ^{insurance}	X ^{jeans}	X ^{jeans}	
Pleasant/unpleasant		X			X ^{insurance}	X ^{insurance}	
Informative/uninformative							
Favorable/unfavorable							X ^{coffee}
B: A _{Brand} (MI and SI) → P _I _{Brand} (MI)							
	Item selection						
	Expert raters	Naive raters	Highest ITC	Highest loading (PAF)	Highest loading (CFA)	Highest SMC (CFA)	Rasch
A_{Brand} MI (baseline)							
Good/bad				X ^{coffee}			
Like/dislike	X		X ^{jeans, pain relievers, coffee}	X ^{jeans, pain relievers}	X ^{jeans, pain relievers, coffee}	X ^{jeans, pain relievers, coffee}	X ^{jeans, pain relievers, coffee}
Pleasant/unpleasant		X	X ^{insurance}	X ^{insurance}	X ^{insurance}	X ^{insurance}	X ^{insurance}
Useful/useless							

ITC: item-to-total-correlation; PAF: principal axis factoring; CFA: confirmatory factor analysis; SMC: squared multiple correlation.

Overall, the Rasch analysis yields highly similar results to the classical test-theory-based criteria. Rasch only suggests a different item (*favorable/unfavorable*) for A_{Ad} in the coffee category. In all the other product categories, the Rasch model and the classical test theory-based approaches identify the same SI.

The results in Table 3 further show that the SI selection patterns of statistical (i.e., classical test theory- and Rasch-based) approaches overlap considerably with the judgmental selection patterns (albeit less so when naive raters provide an assessment). This finding is interesting, because statistical approaches take data idiosyncrasies across different product categories into account, whereas judgment-based approaches base item selection solely on content/face validity considerations (i.e., without access to the data).

In summary, broadly speaking, the different selection approaches seem to converge. The question is whether the selection approaches converge toward the *best* item (i.e., the item with the highest predictive validity).

3.3. Predictive validity of selected single items

Following prior studies in the field (Bergkvist & Rossiter, 2007, 2009; Diamantopoulos et al., 2012), the predictive validity assessment first builds on a baseline model, which captures the correlation between the MI scales of the criterion and predictor constructs based on the average of the item scores (see also Rossiter, 2002). The next step of the analysis involves computing the correlation coefficient between each individual item of the predictor construct (as well as the global, tailor-made SI) and the MI scale of the criterion construct. In a third step, the analysis compares the resulting correlation coefficients with the baseline model's coefficient. This comparison allows for identifying the best SI; that is, the item that displays the highest correlation with the criterion construct.

Table 4 shows the number of instances each approach selects the best performing SI in terms of predictive validity in each product category, as well as overall. The first column in the table shows the number

Table 4

Efficacy of selection approaches in choosing the best SI.

Construct/product category	Number of SIs performing not significantly different than MI ($\alpha = 0.05$)	Global SI performing not significantly different than MI ($\alpha = 0.05$)	Expert raters	Naive raters	Highest ITC	Highest loading (PAF)	Highest loading (CFA)	Highest SMC (CFA)	Rasch	Sum across methods	Average across methods
A _{Ad} (5 items)											
Insurance	0	0	0	0	1	1	0	0	0	2	0.29
Jeans	0	1	1	0	1	1	0	0	1	4	0.57
Pain relievers	1	0	1	0	1	1	1	1	1	6	0.86
Coffee	0	0	0	0	0	0	0	0	0	0	0.00
Hit rate	0.05 ^a	0.25	0.50	0.00	0.75	0.75	0.25	0.25	0.50		
A _{Brand} (4 items)											
Insurance	0	0	1	0	0	0	0	0	0	1	0.14
Jeans	0	0	0	1	0	0	0	0	0	1	0.14
Pain relievers	2	1	0	1	0	0	0	0	0	1	0.14
Coffee	0	1	0	0	0	1	0	0	0	1	0.14
Hit rate	0.13 ^a	0.50	0.25	0.50	0.00	0.25	0.00	0.00	0.00		
Number of successes	3	3	3	2	3	4	1	1	2		
Total hit rate	0.08 ^a	0.38 ^b	0.38	0.25	0.38	0.50	0.13	0.13	0.25		

^a Average value is based on the total number of comparisons (i.e., 20 for A_{Ad}, 16 for A_{Brand} and 36 for total).^b Average value is based on the total number of comparisons of the global SI (i.e., 8).

of items with comparable validity to the MI scale of A_{Ad} and A_{Brand} in each product category; for example, in the pain relievers category, two items perform just as well as the MI scale of A_{Brand} , whereas in the jeans category, all the individual items perform significantly worse. The second column in the table compares the predictive validity of the global (tailor-made) SI with that of the corresponding MI scales; for example, the global SI performs just as well as the MI scale of A_{Ad} in the jeans category (and thus outperforms the individual items comprising the A_{Ad} scale, as none of them attain the full scale's predictive validity). The next seven columns show the extent to which each selection approach manages to identify the best performing SI. A value of 1 denotes that the approach succeeds in identifying the best item in a certain construct/product category combination, while a 0 marks a failure. This information facilitates the computation of hit rates, or the percentage of selections in which each approach identifies the SI with the highest predictive validity.

In terms of A_{Ad} , all the approaches identify the optimal SI in at least two instances, with the exception of the naive raters (who fail to identify the best item) and the CFA criteria (which succeed only once). The comparison of the approaches' performance across product categories shows considerable variation. For example, in the pain relievers category, most approaches identify the best item, whereas no approach identifies the optimal SI in the coffee category. Concerning A_{Brand} , the approaches generally fail to identify the best SI as mirrored in their low average hit rates. Specifically, ITCs, CFA-based criteria, and the Rasch approach always suggest other items than the best SI. Only the naive raters succeed in indicating the best SI in two of the four cases.

In terms of overall performance, loadings from PAF achieve the highest hit rate (0.50), followed by ITCs, expert raters and the global (tailor-made) item (0.38). In contrast, both CFA-based criteria clearly lag behind the other approaches with hit rates of merely 0.13.

4. Discussion and conclusions

This study extends prior literature by examining whether expert raters identify the doubly concrete nature of constructs that previous research based on the C-OAR-SE procedure presents as exemplary in this respect. The results clearly show that *Rossiter's* (2002) as well as *Bergkvist and Rossiter's* (2007, 2009) arguments regarding the concrete nature of A_{Ad} and A_{Brand} are not mirrored in experts' opinions on the adequacy of SI measures for capturing these constructs. More recently, *Rossiter* (2011) even suggested that *all* constructs can be broken down into concrete entities, which implies that any construct can be adequately measured with a single item or aggregations thereof. "Even further-reaching in C-OAR-SE theory is the realization that all 'complex' or 'abstract' constructs (...) are no more than aggregations of 'doubly concrete' constructs (beliefs mostly) and these are each measured uncontestedly with a single item" (*Rossiter*, 2011, p. 1567). However, this study's results cast serious doubt on *Rossiter's* (2011) notion in at least two respects. First, as previously noted, there can be vast disagreement among expert raters as to what makes a construct doubly concrete even with regard to relatively simple constructs. This disagreement potentially stems from different bases of judgments that raters draw upon in their assessment of objects and attributes (*Diamantopoulos*, 2005). For example, while some expert raters may consider an object's denotative meaning, others may base their assessment on its connotative meaning. Second, single items clearly lag behind their multi-item counterparts in terms of predictive validity (see also *Diamantopoulos et al.*, 2012). This result, together with the multitude of conceptual and practical concerns regarding single item use—which are well documented in prior literature (e.g., *Fuchs & Diamantopoulos*, 2009, *Gardner, Cummings, Dunham, & Pierce*, 1998, *Kwon & Trail*, 2005, *Loo*, 2002, *Nagy*, 2002, *Sarstedt & Wilczynski*, 2009)—casts further doubt on the notion that virtually any construct can be adequately measured with single items.

Further analysis of SI selection results shows that the majority of approaches converge broadly. Specifically, classical test theory- and Rasch-based approaches generally exhibit consistent selection patterns across all combinations of constructs/items and product categories. From a practical perspective, researchers may be tempted to conclude that high levels of consistency among the statistical approaches also indicate that the identified SI is optimal in terms of predictive validity. However, the results are not encouraging in this respect. Specifically, the low total hit rates ranging between 13% (CFA-based approaches) and 50% (loadings from PAF) do not provide real confidence in *any* of the approaches, making SI choice a (very) risky task. Even with the best-case scenario (based on PAF), few researchers would be willing to accept a 50:50 chance that their SI selection will provide them with the best item. Furthermore, the results show that expert and naive raters' assessment of the items' suitability may vary considerably (*Table 2*).

Practical considerations associated with the implementation of the various approaches to SI selection further exacerbate the discouraging picture that the low hit rates paint. Most obviously, relying on any of the classical test theory- or Rasch-based approaches results in the paradox that an informed SI selection requires the administration of the full MI scale, unless a well-documented scale development report and sufficient evidence of prior scale use provide the required statistical information. However, even if prior research can supply such information, the question arises whether one should accept and rely on the information without validating the measure in the specific data set at hand. Irrespective of whether one is dealing with an MI or an SI measure, if such a measure worked in the past, this does not automatically guarantee that the measure will work in the future. Importantly, whereas with an MI scale one can readily evaluate its psychometric properties in *any* data set, the same does not apply to an SI measure. In the latter case, the assessment of, say, reliability and convergent validity necessitates access to the full scale from which researchers draw the SI (*Bergkvist*, 2015). Similarly, applying standard procedures to assess discriminant validity is not possible if researchers measure one of the constructs involved with an SI measure (e.g., *Fornell & Larcker*, 1981; *Henseler, Ringle, & Sarstedt*, 2015).

Given the above problems, researchers may opt for a global (tailor-made) SI. The use of such self-standing SIs rests on the assumption that all respondents automatically consider different aspects of the construct (*Fuchs & Diamantopoulos*, 2009). This, however, assumes that respondents are aware of the full scope of the construct's conceptual domain, a situation unlikely to occur in research practice, even when researchers provide the respondents with an accurate description of the construct's nature (*Sackett & Larson*, 1990). In addition, no methodological guidelines exist on how to go about constructing (and validating) a global SI measure *without*, at the same time, having access to an MI measure of the same construct. At most, one could apply a Rasch model to test the fit of the global item; however, this again implies additional data collection in the form of a pre-study.

Another option is to rely on expert raters to ex ante identify the best SI, which *Rossiter* (2002, 2011) recommends. However, apart from the considerable variability in their assessments (see *Table 2* earlier), the experts' choice may be hard to defend as their opinion essentially implies a *hypothesis* about an SI but does not represent empirical evidence that this item is suitable. As this study's results show, expert raters often fail to identify the best SI (as is the case, e.g., with A_{Brand} in the jeans, pain relievers, and coffee categories—see *Table 4*). Only (ex post) empirical assessment allows for revealing such failure. To make matters worse, expert judgment on whether or not an SI is sufficient in the first place, proves largely uninformative. For example, in respect of A_{Brand} , the global SI performs similarly to the MI scale in two of the four product categories (*Table 4*). And yet, only *one* of the 13 experts deems an SI appropriate for measuring A_{Brand} (*Table 2*). This result reinforces the need for empirical evidence either based on the classical test theory, or on the Rasch model, especially since the best SI depends not

only on the focal construct, but also on the product category under consideration (and—as becomes evident when using the Rasch model—also on the distribution of respondents). Researchers can hardly expect experts to take these intricacies accurately into account when choosing an SI. The lack of stability regarding the selection of the best SI, due to its dependency on the product category, also curtails the possible use of known MI scale properties, such as item factor loadings, or ITCs.

Finally, the question arises whether researchers should combine the different SI selection approaches in order to offset their weaknesses, even though this step implies increased time and costs and, thus, rather runs contrary to the spirit of using SI measures. However, owing to the of approaches' high level of correspondence, their combination cannot substantially improve the SI selection task. For example, all approaches perform equally poorly in terms of A_{Ad} for coffee (see Table 4).

In summary, even if researchers *knowingly* accept that SI measures are likely to lag behind their MI counterparts in terms of predictive validity, they still have to surmount the challenge of identifying an appropriate SI. As their poor hit rates show, neither rater assessments nor statistical criteria prove valuable in this respect; using a global (tailor-made) SI does not offer a viable alternative either.

5. Limitations and further research

While this study is the first to empirically address the issues of identifying the doubly concrete nature of constructs and selecting the best SI

from a set of competing items, its design is subject to limitations, which future research should address. First, and most obvious, replications of the present study using other supposedly doubly concrete constructs and product categories are necessary to cross-validate this study's findings in different settings. Second, although highly desirable in terms of internal validity, restricting the current investigation to constructs that become applicable only during the study (attitude toward hitherto unknown ads and brands) inevitably limits the external validity. Further research should therefore address the issue of SI selection taking established attitudes (i.e., with real brands and/or adverts) into account. Third, while predictive validity is without doubt important, this validity type is not the final word in terms of measurement features. Therefore, future studies should also consider other measurement properties, most notably test–retest reliability. Fourth, given that the expert opinions are overwhelmingly *against* the use of an SI to measure what Rossiter (2002, 2011) clearly denotes as doubly concrete constructs, future research should identify means for assessing how concrete or abstract a particular construct is. Last, but by no means least, qualitative insights on *why* expert raters deem some items more suitable than others, constitute another avenue for future research. Relating the justification of the expert decision to the suitability of the suggested SI could possibly lead to an argument-based selection procedure that would allow for distinguishing between trustworthy and equivocal judges. Such a procedure should hopefully result in (much) higher hit rates and, hence, a more reliable mechanism for SI selection than is currently the case.

Appendix A. Rasch model for polytomous responses

Item response theory (e.g., Singh, 2004) and especially the Rasch model (Andrich, 1988; Rasch, 1960), on which marketing research studies increasingly draw, are alternative frameworks for construct measurement (e.g., Salzberger & Koller, 2013; Salzberger, Newton & Ewing, 2014). The separation of item and person characteristics with respect to observed responses is one of the Rasch model's most important properties. Item and person measures, as well as the outcomes of measurement, locate items and persons on the same latent continuum (e.g., Salzberger, 2009). While the person measures have an equivalent in the factor scores in the classical test theoretic framework, no direct counterpart exists for the item measures in classical test theory. This study adopts the polytomous Rasch model (e.g., Andrich, 1978) because the response scale comprises seven categories.

Eq. (A.1) shows the probability that the response a_{vi} of respondent v to item i equals the score x (ranging from 0 to m), given the person location β_v , the item overall location δ_i , and the centralized threshold locations τ_{ij} of j thresholds (Andrich, 1988). The denominator γ is the sum of the numerators across all response categories. The numerator for the lowest category (0) is 1:

$$P(a_{vi} = x | \beta_v, \tau_{ij}, j = 1 \dots m, 0 < x \leq m) = \frac{e^{\left(\sum_{j=1}^x -\tau_{ij}\right) + x(\beta_v - \delta_i)}}{\gamma}, \text{ with} \quad (A.1)$$

$$\gamma = 1 + \sum_{k=1}^m e^{\left(\sum_{j=1}^k -\tau_{ij}\right) + k(\beta_v - \delta_i)}$$

Eq. (A.2) shows the information of a polytomous item i based on the Rasch model (Linacre, 2005). For equivalent formulae, see Dodd, De Ayala, & Koch (1995), and Muraki (1993). $E(\beta)$ is the expected score for a person with the location β , k is the response category scored from 0 to m , and $P_k(\beta)$ is the probability of response category k given β :

$$I_i(\beta) = \sum_{k=0}^m (k - E(\beta))^2 P_k(\beta) \quad (A.2)$$

In a Rasch analysis, the choice of an optimal SI depends on the matching of a polytomous item's threshold estimates (i.e., the boundaries between adjacent answer categories on the latent continuum) and the respondent distribution (i.e., the location of the respondents' measures on the same latent continuum). For example, a seven-category item requires the estimation of six thresholds that divide the latent continuum into seven sections. An ideal item would yield thresholds that separate the person distribution into seven groups of equal size. The best item therefore minimizes the absolute differences between the item thresholds and the cutoff points of seven classes with each containing one seventh of the total sample of respondents.

Appendix B. Rasch item statistics

In Table B1, the first number states the overall item location (mean of thresholds). The second value represents the fit residual statistic (large negative values indicate over-discrimination, large positive values imply under-discrimination). The third value shows the p -value of the chi-square test

of fit (e.g., Salzberger, 2009). Table B2 shows the average distance of the actual thresholds from their ideal locations for all items considered as SI measures after the fit assessment.

Table B1
Rasch Item Statistics.

	Insurance	Jeans	Pain relievers	Coffee
Overall item location/fit residual statistic/p-value				
<i>A_{Ad}</i> ^a				
Like/dislike	0.21/–1.23/0.58	0.23/–1.93/0.73	0.16/–2.22/0.20	–0.06/–2.22/0.51
Good/bad	–0.12/0.29/0.66	–0.22/–0.44/0.76	–0.22/0.45/0.32	0.05/0.84/0.29
Favorable/unfavorable	0.19/1.95/0.27	–0.16/2.50/0.43	0.46/3.35/0.16	0.14/2.48/0.54
Global SI	–0.28/–1.36/0.04	0.15/–0.85/0.34	–0.40/–1.08/0.03	–0.13/–1.55/0.14
<i>A_{Brand}</i> ^b				
Good/bad	0.46/0.46/0.47	–0.13/1.45/0.66	–0.29/0.06/0.85	–0.10/–0.70/0.45
Like/dislike	–0.02/–1.99/0.67	0.15/–2.33/0.56	0.33/–0.13/0.68	0.14/–1.32/0.85
Pleasant/unpleasant	–0.28/–1.77/0.78	–0.01/–2.34/0.39	0.18/–1.24/0.35	–0.34/1.08/0.50
Global SI	–0.15/1.36/0.70	–0.03/1.18/0.93	–0.21/1.46/0.13	0.31/1.86/0.06

^a Pleasant/unpleasant and informative/uninformative are not considered due to lack of fit.

^b Useful/useless is not considered due to lack of fit.

Table B2
Average distance of the actual thresholds from the ideal locations.

	Insurance	Jeans	Pain relievers	Coffee
<i>A_{Ad}</i> ^a				
Like/dislike	1.34	0.77	0.85	1.49
Good/bad	1.67	1.39	0.91	1.61
Favorable/unfavorable	1.53	1.16	0.89	1.14
Global SI	1.41	1.43	1.12	1.65
<i>A_{Brand}</i> ^b				
Good/bad	3.06	2.12	2.19	1.72
Like/dislike	1.72	1.35	1.36	1.26
Pleasant/unpleasant	1.31	1.45	1.36	1.53
Global SI	2.50	2.77	2.81	2.18

The best item, which minimizes the discrepancy, is printed in bold.

^a Pleasant/unpleasant and informative/uninformative are not considered due to lack of fit.

^b Useful/useless is not considered due to lack of fit.

Note that while the analysis assumes that the person distribution is normal, in principle, the analysis could also use the actual distribution of respondent measures. This approach would, however, be solely data driven and involves the risk of capitalizing on chance. When facing a new application of the scale, the exact shape of the distribution is usually unknown. A normal distribution seems to be a plausible default expectation (Salzberger, 2009). Note that in CFA, a normal distribution of measures is typically also assumed.

References

- Aaker, D. A., Kumar, V., & Day, G. S. (2010). *Marketing research*. New York, NY: John Wiley & Sons.
- Andrich, D. (1978). A rating formulation for ordered response categories. *Psychometrika*, 43(4), 561–573.
- Andrich, D. (1988). *Rasch models for measurement*. Sage University Series on quantitative applications in social sciences. Newbury Park, CA: Sage Publications.
- Andrich, D., Sheridan, B. S., & Luo, G. (2012). *Rumm2030: Rasch unidimensional measurement models (software)*. Perth, Western Australia: RUMM Laboratory.
- Bergkvist, L. (2015). Appropriate use of single-item measures is here to stay. *Marketing Letters*, 26(3), 237–243.
- Bergkvist, L., & Rossiter, J. R. (2007). The predictive validity of multiple-item versus single-item measures of the same constructs. *Journal of Marketing Research*, 44(2), 175–184.
- Bergkvist, L., & Rossiter, J. R. (2009). Tailor-made single-item measures of doubly concrete constructs. *International Journal of Advertising*, 28(4), 607–621.
- Christophersen, T., & Konradt, U. (2011). Reliability, validity, and sensitivity of a single-item measure of online store usability. *International Journal of Human-Computer Studies*, 69(4), 269–280.
- Crocker, L., & Algina, J. (2008). *Introduction to classical and modern test theory*. Mason, OH: Cengage.
- De Boer, A. G. E. M., van Lanschot, J. J. B., de Haes, J. C. J. M., & Sprangers, M. A. G. (2004). Is a single item visual analogue scale as valid, reliable, and responsive as multi-item scales in measuring quality of life? *Quality of Life Research*, 13(2), 311–320.
- DeMoranville, C. W., & Bienstock, C. C. (2003). Question order effects in measuring service quality. *International Journal of Research in Marketing*, 20(3), 217–231.
- DeVellis, R. F. (2003). *Scale development: Theory and applications*. Thousand Oaks, CA: Sage.
- Diamantopoulos, A. (2005). The C-OAR-SE procedure for scale development in marketing: A comment. *International Journal of Research in Marketing*, 22(1), 1–9.
- Diamantopoulos, A., Sarstedt, M., Fuchs, C., Wilczynski, P., & Kaiser, S. (2012). Guidelines for choosing between multi-item and single-item scales for construct measurement: A predictive validity perspective. *Journal of the Academy of Marketing Science*, 40(3), 434–449.
- Diamantopoulos, A., Siguaw, J. A., & Cadogan, J. W. (2008). Measuring abstract constructs in management and organizational research: The case of export coordination. *British Journal of Management*, 19(4), 389–395.
- Dodd, B. G., De Ayala, R. J., & Koch, W. R. (1995). Computerized adaptive testing with polytomous items. *Applied Psychological Measurement*, 19(1), 5–22.
- Dolbier, C. L., Webster, J. A., McCalister, K. T., Mallon, M. W., & Steinhardt, M. A. (2005). Reliability and validity of a single-item measure of job satisfaction. *American Journal of Health Promotion*, 19(3), 194–198.
- Drolet, A. L., & Morrison, D. G. (2001). Do we really need multiple-item measures in service research? *Journal of Service Research*, 3(3), 196–204.
- Fleiss, J. L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5), 378–382.
- Fornell, C. G., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50.
- Fuchs, C., & Diamantopoulos, A. (2009). Using single-item measures for construct measurement in management research. *Business Administration Review*, 69(2), 195–210.
- Gardner, D. G., Cummings, L. L., Dunham, R. B., & Pierce, J. L. (1998). Single-item versus multiple-item measurement scales: An empirical comparison. *Educational and Psychological Measurement*, 58(6), 898–915.
- Hair, J. F., Sarstedt, M., Ringle, C. M., & Mena, J. A. (2012). An assessment of the use of partial least squares structural equation modeling in marketing research. *Journal of the Academy of Marketing Science*, 40(3), 414–433.
- Henseler, J., Ringle, C., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43(1), 115–135.

- Kamakura, W. A. (2015). Measure twice and cut once: The carpenter's rule still applies. *Marketing Letters*, 26(3), 237–243.
- Keiningham, T. L., Cooil, B., Aksoy, L., Andreassen, T. W., & Weiner, J. (2007). The value of different customer satisfaction and loyalty metrics in predicting customer retention, recommendation, and share-of-wallet. *Managing Service Quality*, 17(4), 361–384.
- Kuppelwieser, V., & Sarstedt, M. (2014a). Exploring the influence of customers' time horizon perspectives on the satisfaction–loyalty link. *Journal of Business Research*, 67(12), 2620–2627.
- Kuppelwieser, V., & Sarstedt, M. (2014b). Applying the future time perspective scale to advertising research. *International Journal of Advertising*, 33(1), 113–136.
- Kwon, H., & Trail, G. (2005). The feasibility of single-item measures in sport loyalty research. *Sport Management Review*, 8(1), 69–88.
- Linacre, J. M. (2005). Dichotomous and polytomous category information. *Rasch Measurement Transactions*, 19(1), 1005–1006.
- Loo, R. (2002). A caveat on using single-item versus multiple-item scales. *Journal of Managerial Psychology*, 17(1), 68–75.
- MacKenzie, S. B., & Lutz, R. J. (1989). An empirical examination of the structural antecedents of attitude toward the ad in an advertising pretesting context. *Journal of Marketing*, 53(2), 48–65.
- Mitchell, A. A. (1986). The effect of verbal and visual components of advertisements on brand attitudes and attitude toward the advertisement. *Journal of Consumer Research*, 13(1), 12–24.
- Muraki, E. (1993). Information functions of the generalized partial credit model. *Applied Psychological Measurement*, 17(4), 351–363.
- Nagy, M. S. (2002). Using a single-item approach to measure facet job satisfaction. *Journal of Occupational & Organizational Psychology*, 75(1), 77–86.
- Netemeyer, R. G., Bearden, W. O., & Sharma, S. (2003). *Scaling procedures: Issues and applications*. Thousand Oaks, CA: Sage Publications.
- Nunnally, J. C., & Bernstein, I. H. (1994). *Psychometric theory* (3rd ed.). New York, NY: McGraw-Hill.
- Rasch, G. (1960). *Probabilistic models for some intelligence and attainment tests*. Copenhagen: Danish Institute for Educational Research (reprint 1980, expanded ed. with foreword and afterword by B.D. Wright). Chicago, IL: The University of Chicago Press.
- Revelle, W. (1979). Hierarchical cluster analysis and the internal structure of tests. *Multivariate Behavioral Research*, 14(1), 57–74.
- Reynolds, F. D., & Darden, W. R. (1971). Mutually adaptive effects of interpersonal communication. *Journal of Marketing Research*, 8(4), 449–454.
- Rossiter, J. R. (2002). The C-OAR-SE procedure for scale development in marketing. *International Journal of Research in Marketing*, 19(4), 305–335.
- Robins, D. L., Fein, D., Barton, M. L., & Green, J. A. (2001). The modified checklist for autism in toddlers: An initial study investigating the early detection of autism and pervasive developmental disorders. *Journal of Autism and Developmental Disorders*, 31(2), 131–144.
- Rossiter, J. R. (2011). *Measurement for the social sciences. The C-OAR-SE method and why it must replace psychometrics*. Berlin: Springer.
- Rossiter, J. R., Percy, L., & Donovan, R. J. (1991). A better advertising planning grid. *Journal of Advertising Research*, 31(5), 11–21.
- Ruekert, R. W., & Churchill, G. A. (1984). Reliability and validity of alternative measures of channel member satisfaction. *Journal of Marketing Research*, 21(2), 226–233.
- Sackett, P. R., & Larson, J. R. (1990). Research strategies and tactics in industrial and organizational psychology. In M. D. Dunnette, P. L. Ackerman, & L. M. Hough (Eds.), *Handbook of industrial and organizational psychology* (2nd ed.) Vol. 1. (pp. 419–489). Palo Alto, CA: Consulting Psychology Press.
- Salzberger, T. (2009). *Measurement in marketing research: An alternative framework*. Cheltenham, UK: Edward Elgar.
- Salzberger, T., & Koller, M. (2013). Towards a new paradigm of measurement in marketing. *Journal of Business Research*, 66(9), 1307–1317.
- Salzberger, T., Newton, F. J., & Ewing, M. T. (2014). Detecting gender item bias and differential manifest response behavior: A Rasch-based solution. *Journal of Business Research*, 67(4), 598–607.
- Sarstedt, M., & Wilczynski, P. (2009). More for less? A comparison of single-item and multi-item measures. *Business Administration Review*, 69(2), 211–227.
- Sarstedt, M., Wilczynski, P., & Melewar, T. C. (2013). Measuring reputation in global markets—A comparison of reputation measures' convergent and criterion validities. *Journal of World Business*, 48(3), 329–339.
- Singh, J. (2004). Tackling measurement problems with item response theory: Principles, characteristics, and assessment, with an illustrative example. *Journal of Business Research*, 57(2), 184–208.
- Stanton, J. M., Sinar, E. F., Balzer, W. K., & Smith, P. C. (2002). Issues and strategies for reducing the length of self-report scales. *Personnel Psychology*, 55(1), 167–194.
- Wanous, J. P., Reichers, A. E., & Hudy, M. J. (1997). Overall job satisfaction: How good are single-item measures? *Journal of Applied Psychology*, 82(2), 247–252.
- Youngblut, J. M., & Casper, G. R. (1993). Single-item indicators in nursing research. *Research in Nursing and Health*, 16(6), 459–465.